d2025 Division C Tournament - Univ. Of Georgia

Materials Science Lab Written Exam

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School Name/Team: ____

Student Names: ____

1. There will be a dry lab portion for this exam.
2. You are allowed one unique 8.5 x 11” front and back cheat sheets as well as one calculator of any class.
3. This test will be scored out of 86 points. Point values for each question will be listed.

Instructions:

<u>Multiple Choice</u> <u>(15 Questions; 15 Points)</u>

1.What relationship exists between the size of a quantum dot and the wavelength that it absorbs?

a. Smaller size associates with longer wavelengths absorbed.
b. There is no association between the size of a quantum dot and the wavelength absorbed.
c. Larger size associates with longer wavelengths absorbed
d. Quantum dots absorb all light regardless of size

2. Covid 19 home tests contained incredibly small spheres of gold known as colloidal gold. These nanoparticles would bond to antibodies that recognized the virus and would conglomerate together to show a result. What kind of nanomaterial is gold?

a. Metal nanomaterial
b. One dimensional nanomaterial
c. Metal-Oxide nanomaterial
d. Quantum dot nanomaterial

3. The gold nanoparticles in these tests are actually a nice ruby red color, not gold. This is also the same case for a wide set of nanomaterials. What phenomenon is responsible for the shift in colors of the gold at the nano scale?

a. Localized Surface Plasmon Resonance
b. Quantum Confinement
c. Rayleigh Scattering
d. Yang-Rupesh Effect

4. What is the hybridization of graphene/carbon nanotubes?

a. sp2
b. dsp2
c. sp
d. None of the above

5. Naturally occurring 2D nanomaterials are quite hard to come by, which hybridization is the most common for 2D nanomaterials.

a. dsp3
b. dsp
c. sp2
d. sp

6. Amorphous materials are materials that are …

a. Isotropic
b. Anisotropic
c. Crystalline
d. Autotrophic

7. How are the valence and conduction bands in dirac cones within graphene arranged?

a. Overlapping each other at every point.
b. Overlapping each other at 5 distinct points
c. Overlapping each other at 1 point, remaining separate everywhere else
d. They do not overlap each other at all.

8. What is the region within a molecule that excited electrons can enter?

a. Valence band
b. Conduction band
c. Van der Wall band
d. Ionic band

9. How would adding a nitrogen based dopant to a nanomaterial impact its properties?

a. Higher Young’s modulus
b. Increased Conductivity
c. Improved magnetic properties
d. Shift in band gap

10. During the lithography process a photoresist coated silicon wafer has a mask applied to it, creating a desired pattern. What is the next step in the lithography process?

a. Light is shone on the wafer and chemical reactions occur where the mask doesn’t cover the wafer.

b. A chemical surfactant is poured over the wafer, eating away the regions where the mask isn't present.
c. The wafer is left to bake in an industrial grade oven where the pattern on the mask is etched into the wafer.
d. The wafer is dipped into an acid mixture where the acid imprints the mask's imprint onto the wafer.

11. Which of the following is a biomedical application of gold nanoparticles?
    a. Soil fertilizer
    b. Diseases detector
    c. Bone density agent
    d. Water purifier

12. Which of the following techniques cannot be used to determine the nanoparticles size?
    a. Conventional Light Microscope
    b. Transmission Electron Microscope
    c. Scanning Electron Microscope
    d. Dynamic Light Scattering Spectrophotometer

13. Top-down approach in nanoparticle synthesis includes which of the following:
    a. Chemical vapor deposition
    b. Molecular beam epitaxy
    c. Chemical etching
    d. Thermal decomposition

14. Which of the following nanoparticles has magnetic properties?
    a. Titanium dioxide nanoparticles
    b. Cobalt monoxide nanoparticles
    c. Gold nanoparticles
    d. All of the above

15. Which statement correctly describes the effects of decreasing the nanoparticle size?
    a. It increases the surface-area-to-volume ratio, and reduces surface reactivities
    b. It decreases the surface-area-to-volume ratio, and reduces the catalytic activity
    c. It decreases the surface-area-to-volume ratio, and increases the catalytic activity.
    d. It increases the surface area-to-volume ratio, and increases the surface reactivities

Free Response Questions (FRQ)

Ved and His Computer (5 questions; 9 points)

After losing a Valorant game Ved decides that despite going triple negative, his computer was the issue. With this in mind, he decides to go out and make, not buy, his own computer, starting with the CPU.

1. Assuming Ved has all the necessary materials and equipment available, briefly describe the process of photolithography. (Order needs to be correct) **2 points*

2. The beginning stages of photolithography are crucial as selecting different material wafers directly impacts the final product. Ved decides to go with the industry standard: a monocrystalline silicon wafer. What makes this material the industry standard, list 2 properties? (Think about its electrical properties and workability). **2 points*

3. Somewhere down the line, something goes horribly long. During the exposure process, Ved had the correct power of UV light, but he left the wafer exposed for too long. How could this impact the wafer? **2 points*

4. In the final steps of the process, Ved begins to etch his wafer. In this process, Ved uses Hydrofluoric acid. Despite being a weak acid, it remains a mainstay in lithography, why is this? (Think about the typical material for wafers) **1 point*

5. Ved finally has it, a broken CPU, it was never bound to work: photolithography is hard. What makes photolithography special when compared against other methods of lithography? **3 points*

<u>Ved and His Car</u> <u>**(3 questions; 9 points)**</u>

Still in desperate need of a CPU to continue his Valorant career, Ved begins to steal and sell catalytic converters.

1. Catalytic converters sell for very good prices, and for very good reason. The metals that they contain tend to be very expensive. List out 3 common elements found within Catalytic Converters. **3 points*

2. The amount of these metals found in catalytic converters is actually very small, and that is because of their very good surface area to volume ratio.

    a. Quickly define what the surface area to volume ratio is. **1 point*

    b. Derive the surface area to volume ratio of a sphere, a common model for these elements. Use r for radius, S for surface area, and V for volume. **3 points*

3. Nanosized grains that contain high amounts of grain boundaries are preferred in catalytic converters.

    a. Quickly define what grain boundaries are, and draw a small picture of one. **2 points*

    b. What about the high number of grain boundaries makes these nanoparticles good for this application? Think about the hot exhaust gasses that are constantly flowing through the converter. **2 points*

Ved and his Homework (4 questions; 9 points)

Ved must really be famous huh. Nonetheless, for his college materials science class, he is assigned a nanomaterial to collect some information on.

1. What are the 2 most common microscopies used in order to characterize nanomaterials? **1 point*

2. Briefly describe how both microscopies work, list out what makes them different. **3 points*

3. List out what properties both microscopies would be used to find, bullet point format is fine. **3 points*

4. If Ved wanted to study internal crystal defects, which microscopie would he pick? Quickly defend your answer. **2 points*

<u>Math | No Work Necessary</u> <u>(11 questions; 11 points)</u>

(Answers may not be exact, pick the closest option)

1. Given that a Quantum dot absorbs 1.96 eV, what color of light did the Quantum dot absorb.

   a. Blue

b. Green
c. Yellow
d. Red

2. Calculate the surface area to volume ratio of a 13 nm diameter nanoparticle, assume that it can be modeled as a sphere.

a. 0.231 nm^-1
b. 0.325 nm^-1
c. 0.462 nm^-1
d. 3.25 nm^-1

3. Calculate the band gap of a 1000 Kelvin nanoparticle.

a. 0.0521 eV
b. 0.0862 eV
c. 0.115 eV
d. 1.24 eV

4. Given that a 1 newton force is applied over a circular area of 0.03m^2, what is the voltage generated? Assume a piezoelectric coefficient of 5x10^-12 C/N.

a. 1.67 x 10^-13 V
b. 1.79 x 10^-10 V
c. 2.13 x 10^-12 V
d. 1.77 x 10^-19 V

5. Heat is being transported through a single walled carbon nanotube at a rate of 52.5 when a temperature difference of 10K is applied. The tube has a length of 1x10^-6m and an effective cross-sectional area of 1.5x10^-18m^2. Calculate the thermal conductivity of the carbon nanotube.

a. 401 W/mK
b. 2200 W/mK
c. 3500 W/mK
d. 6600 W/mK

6. Calculate the stress on a rectangle beam with a square cross sectional area. The side lengths of the face is 1m, and the force applied is 100N.

a. 100 N/m^2
b. 200N/m^2
c. 150N/m^2
d. 10N/m^2

7. What is the Young's modulus of a material that has a stress of 10 Kpa and a strain of 5?

a. 2 KPA
b. 2Pa
c. 4KPa
d. 2Pa

8. What is the wear volume on a material with a normal load of 100N, sliding distance of 10m, a hardness of 2GPa, and a dimensionless wear coefficient of 5x10^-5?

a. 1.5x10^-10 m^3
b. 2.5x10^-11 m^3
c. 3.25x10^-11m^3
d. 4.5x10^-12m^3

9. Which of the following, after manipulation, is equivalent to the Hall-Petch Relation

a. $\sigma y - \sigma_0 = - k*d$^2
b. $\log(\sigma y) = e$^$(\sigma_0 + k*d^{-1/2})$
c. $\sigma y = \ln(\sigma_0) - \sin(k*d)$
d. $-\sigma y + \sigma_0 = - k*d^{-1/2}$

10. What is the formula for Stress?

a. Force/Area
b. Area/Force
c. It's not this
d. It's not this one either

11. What is Young's modulus of a material with length 1m and cross-sectional area 2cm^2 being stretched 0.7 mm under a tensile load of 3000 N?

a. 1 x 10^9 Pa
b. 2 x 10^10 Pa
c. 2 x 10^9 Pa
d. 1 x 10^10 Pa

Math Free Response Question (FRQ) | (Please Show All Work) |

1. Given a spherical gold nanoparticle has a diameter of 22 nm, density of gold is 19.3 g/cm3, 3 x 10^9 particles in the batch. ***(3 parts; 16 points)***

a) Calculate the mass of a single gold nanoparticle and the total mass of all nanoparticles. **5 points*

b) Calculate the surface-area-to-volume ratio of a single nanoparticle, and the total surface area of the batch. **5 points*

c) Calculate the concentration of nanoparticles dispersed in 30 mL of water. In addition, given that each nanoparticle has 1 x 10^6 gold atoms, calculate the molar concentration of gold atoms in the solution. **6 points*

2. An aluminum rod (E = 70 GPa) has a circular cross-section and is 8.0 m long. It is subjected to a 20 kN tensile force. Answer the following questions. Round all answers to one decimal point. ***(3 parts; 17 points)***

a) Determine minimum diameter in millimeters if the rod must not elongate more than 4.0 mm. Assume the aluminum remains in the elastic region. **9 points*

b) If the rod is manufactured with the minimum safe diameter found in part a), determine the stress in megapascals in the rod under the 20 kN tensile load. **3 points*

c) Suppose the rod is instead made of steel (E = 200GPa) but keeps the same diameter from part a). Under the same 20 kN load, calculate the elongation in millimeters. **5 points*